このスライド の PDF 版

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↓

室ホーム／slide

⓿ V : vector space
$v_0, v_1, \dots, v_n \in V$: vectors.
$\{v_i\}_{i=0}^n \underset{\text{identify}}{\longleftrightarrow} E=\{0,1,\dots,n\}$
$\mathcal{L} := \{\mathrm{Span}(S)\cap E : S\subset V\}$
$(E,\mathcal{L})$: matroid.

Poincaré duality of matroids via star subdivisions (survey)

§ Intro

$n\in\mathbb{Z}_{\ge 0}$, $E:=\{0,1,\dots,n\}$, $\mathcal{L}\subset 2^E$

Def

$M:=(E,\mathcal{L})$ is called a loopless matroid

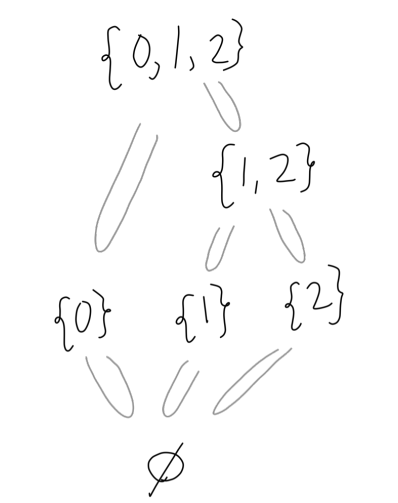
$\overset{\text{def}}{\iff}$ ① $\emptyset, E\in\mathcal{L}$

② $F,G\in\mathcal{L}\Rightarrow F\cap G\in\mathcal{L}$

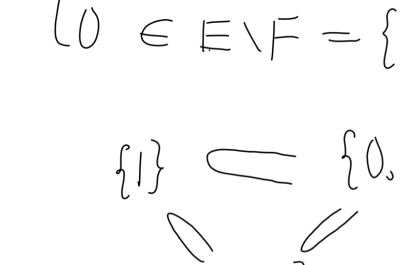
③ $F\in\mathcal{L}$, $e\in E\setminus F$

$\Rightarrow \exists G\in\mathcal{L}$ s.t. $\begin{cases} G \overset{\text{cover}}{\supset} F, \\ e\in G \end{cases}$

$\nexists H\in\mathcal{L}$ s.t. $F\subsetneq H\subsetneq G$



$\begin{cases} F=\{1\} \\ 0\in E\setminus F=\{0,2\} \end{cases}$



- subsets beloning $\mathcal{L}$ are called flats of M
- $\mathcal{L}(M):=\mathcal{L}$ ← the set of flats of M
- $(\mathcal{L}(M),\subset)$ is a lattice
- all maximal chain of flats have the same length, called the rank of M

Example

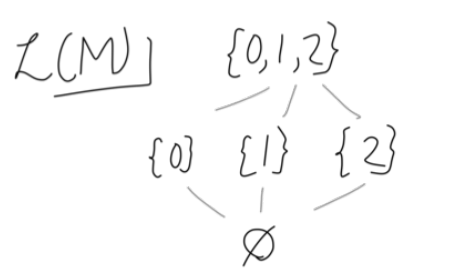
❶ $\mathcal{L}(M):=2^E$
(M is called a Boolean matroid) $B_n:=M$

- $\mathrm{rank}(B_n)=|E|=n+1$

$\emptyset\subsetneq\{0\}\subsetneq\{0,1\}\subsetneq\cdots\subsetneq\{0,1,\dots,n\}=E$

❷ $n=2$, $E=\{0,1,2\}$

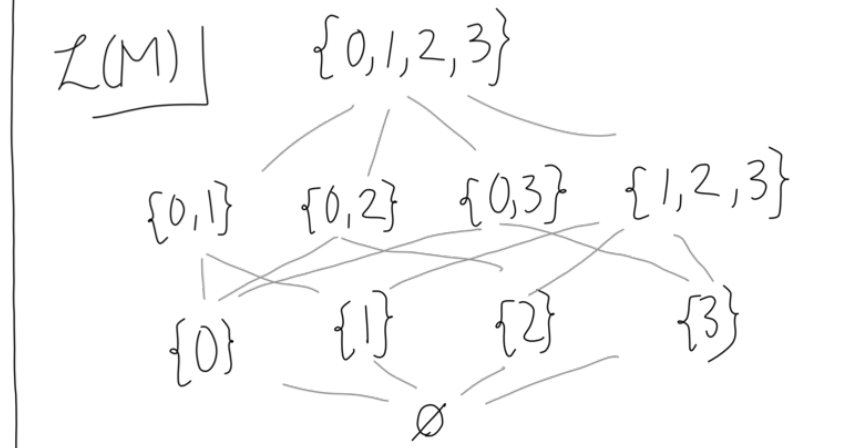
$\mathcal{L}(M)$



- $\mathrm{rank}(M)=2$

❸ $n=3$, $E=\{0,1,2,3\}$

$\mathcal{L}(M)$



- $\mathrm{rank}(M)=3$

- Matroids induces a polyhedral fan

$\bigoplus_{i\in E}\mathbb{Z}e_i$: the free abelian grp on E. ← rank $=n+1$

For $S\subset E$, $e_S:=\sum_{i\in S}e_i$

$N_E:=\dfrac{\bigoplus_{i\in E}\mathbb{Z}e_i}{\mathbb{Z}e_E}$ ← rank n

$N_{E,\mathbb{R}}:=N_E\otimes_{\mathbb{Z}}\mathbb{R}$

Def $\mathcal{P}(M):=\mathcal{L}(M)\setminus\{\emptyset,E\}$

$\Sigma_M:=\{\underset{\substack{\cap\\ N_{E,\mathbb{R}}}}{\mathrm{Cone}(e_F : F\in\mathcal{F})} : \mathcal{F}\underset{\text{flag}}{\subset}\mathcal{P}(M)\}$

(being totally ordered subset w.r.t. $\subset$)

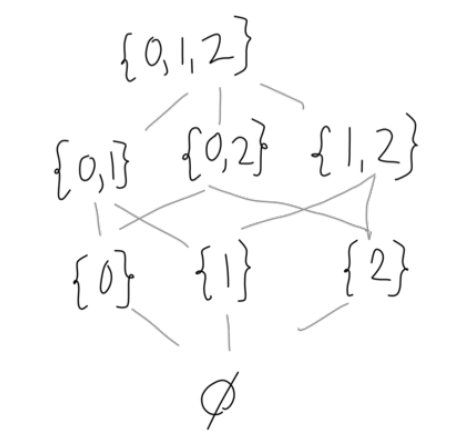
- Σ_M is called the Bergman fan of M
- Σ_M is a unimodular fan of pure dim $=\mathrm{rk}(M)-1$
  ↳ corresp. toric var is smooth

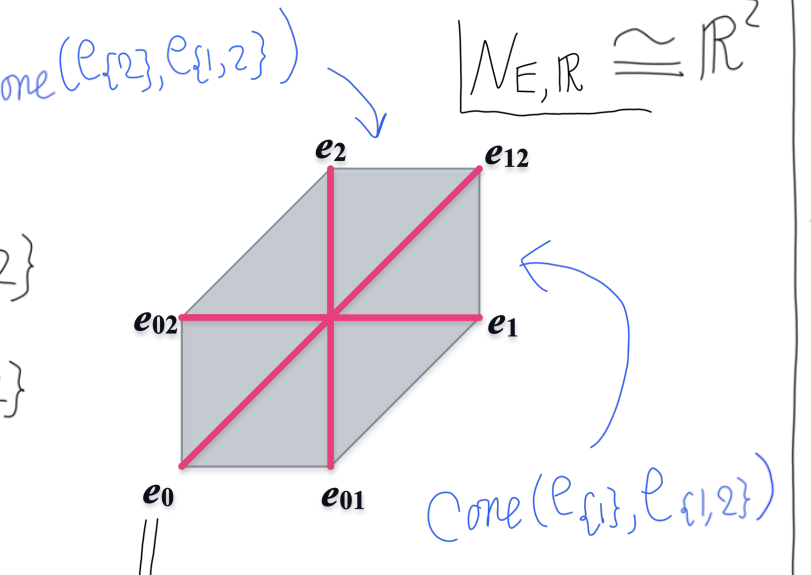
Exa.

❶ $n=2$, B_2: Bloolean $(\mathcal{L}(B_2)=2^{\{0,1,2\}})$

$N_{E,\mathbb{R}}\cong\mathbb{R}^2$

$\mathcal{L}(B_2)$





$\mathrm{rank}(B_2)=3$

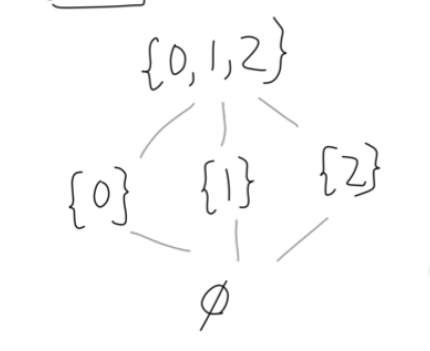
$e_0 = -e_1-e_2$

$\dim\Sigma_{B_2}=2$

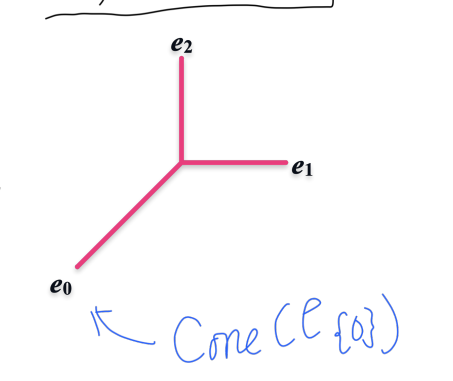
$e_0+e_1+e_2=0$

❷ $n=2$

$\mathcal{L}(M)$

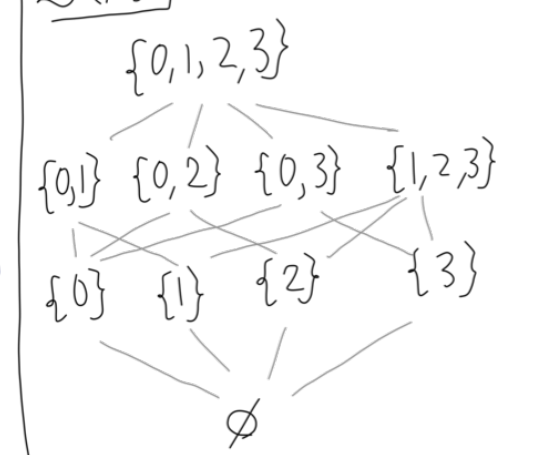


$N_{E,\mathbb{R}}\cong\mathbb{R}^2$



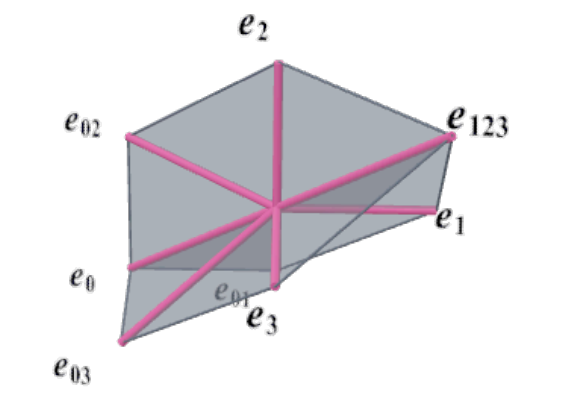
❸ $n=3$

$\mathcal{L}(M)$



$\mathrm{rank}(M)=3$

$N_{E,\mathbb{R}}\cong\mathbb{R}^3$



$\dim\Sigma_M=2$

$X(\Sigma_M):=$ the toric variety $/\mathbb{C}$ corresponding to $\Sigma_M\subset N_{E,\mathbb{R}}$

↳ of dim n, smooth, quasi-projective but non-proper in general

Def

$A^\bullet(M):=A^\bullet(X(\Sigma_M))\cong\dfrac{\mathbb{Z}[x_F]_{F\in\mathcal{P}(M)}}{(\text{some relations})}$

↳ the Chow ring of $X(\Sigma_M)$

called the Chow ring of M

We can study Char poly of M (chrom. poly of a graph) via $A^\bullet(M)$

Thm (Adiprasito-Huh-Katz '18)

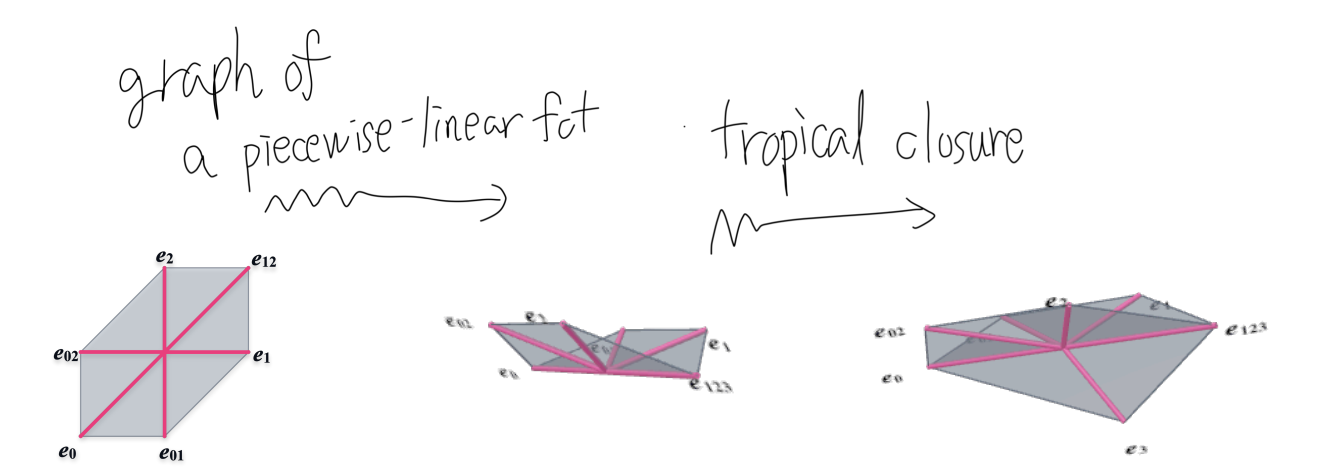
Let M be a loopless matroid on E, rank $=r+1$

$A^\bullet(M)$ satisfies the Kähler package:

- Poincaré duality of dim r ← We talk today
- hard Lefschetz property
- Hodge-Riemann relation

We do not talk today

- $\ell>r\Rightarrow A^\ell(M)=0$
- $A^r(M)\cong\mathbb{Z}$
- if $\ell\le r$,
  $A^\ell(M)\times A^{r-\ell}(M)\longrightarrow A^r(M)\cong\mathbb{Z}$
  $(x,y)\longmapsto xy$
  is a perfect pairing



§ Proof of Thm

History: AHK'18 $\underset{\text{another method}}{\rightsquigarrow}$ Amini-Piquerez'20 $\underset{\text{simplify}}{\rightsquigarrow}$ Hsiao'25

- AHK'18 method: induction on n & r ($n = |E|-1$, $r = \mathrm{rank}(M)-1$)
  matroidal flip
- Amini-Piquerez '20 method: induction on n
  (I) tropical modification (Deletion-Contraction of Matroids)
  + (II) Weak factorization thm for smooth toric var (Blowup center?)
  + (III) Keel's lemma for PD
- Hsiao'25 method:
  Construct an explicit sequence of Blowups,
  in particular, explicit Blowup centers

Below, we may assume that M is simple, i.e., $\forall a\in E$, $\{a\}\in\mathcal{L}(M)$

$(\because M\underset{\text{simplify}}{\rightsquigarrow}\tilde{M}$, $A^\bullet(M)\underset{\text{fact}}{\cong}A^\bullet(\tilde{M}))$

(I) Tropical modification, Deletion-Contraction

Tropical analogue of taking the graph of an algebraic function

$a\in E$: fix

$M\setminus a$: deletion by a

$\mathcal{L}(M\setminus a):=\{F\setminus\{a\} : F\in\mathcal{L}(M)\}$

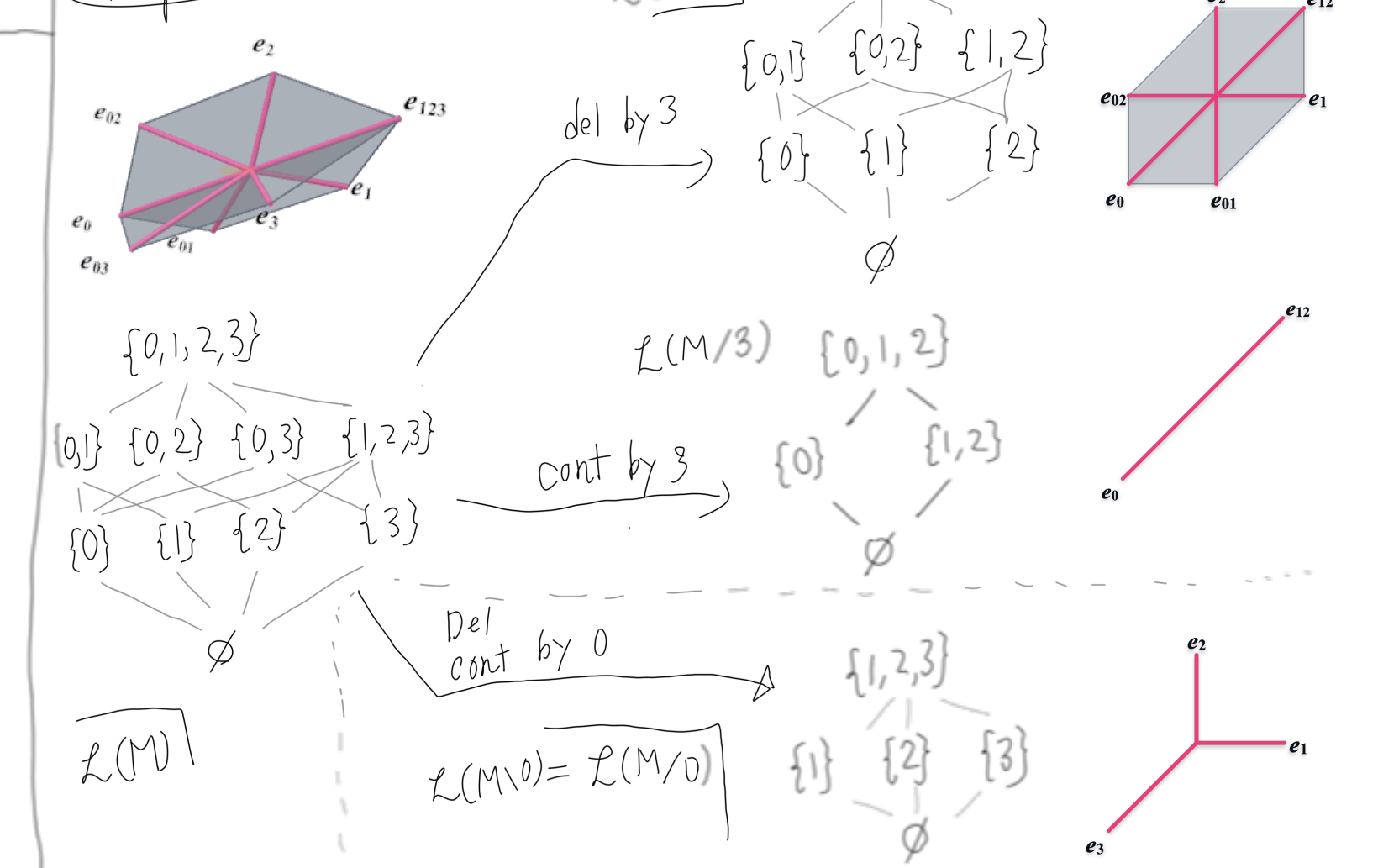
M/a : contraction by a

$\mathcal{L}(M/a):=\{F\setminus\{a\} : F\in\mathcal{L}(M), a\in F\}$

- $M\setminus a$, M/a : loopless matroids on $E\setminus\{a\}$ ← $(\because \{a\}\in\mathcal{L}(M))$
- $\Sigma_{M/a}\underset{\text{subfan}}{\subset}\Sigma_{M\setminus a}$

- $\bigoplus_{i\in E}\mathbb{Z}e_i\twoheadrightarrow\bigoplus_{i\in E\setminus\{a\}}\mathbb{Z}e_i\ \underset{\text{induce}}{\rightsquigarrow}\ N_E\twoheadrightarrow N_{E\setminus\{a\}}\ \underset{\text{induce}}{\rightsquigarrow}\ \Sigma_M\longrightarrow\Sigma_{M\setminus a}$

Example



$a\in E$ is called a coloop $\overset{\text{def}}{\iff}$ $\mathrm{rank}(M)>\mathrm{rank}(M\setminus a)$
(i.e., $\dim\Sigma_M>\dim\Sigma_{M\setminus a}$)

Previous example

1, 2, 3 are non-coloop.

0 is coloop.

FACT

If $\forall a\in E$ is a coloop, then $M=B_n$ (Boolean)

- $X(\Sigma_{B_n})$ is smooth projective,
  $\underset{\text{fact}}{\rightsquigarrow}$ $A^\bullet(B_n)\cong H^{2\bullet}(X(\Sigma_{B_n}))$
  $\rightsquigarrow$ PD of $A^\bullet(B_n)$: OK
- We assume that $M\neq B_n$
- We fix $a\in E$: non-coloop

def $S\subset E$

$\mathrm{cl}_M(S):=\bigcap\{F\in\mathcal{L}(M) : S\subset F\}$

↳ closure of S in M

Consider piecewise-linear map

$$\begin{array}{ccc}\varphi : |\Sigma_{M\setminus a}| & \longrightarrow & N_{E,\mathbb{R}}\\ \cup & & \cup\\ e_G & \longmapsto & e_{\mathrm{cl}_M(G)}\end{array}$$

$(G\in\mathcal{L}(M\setminus a))$

$$e_{\mathrm{cl}_M(G)} = \begin{cases} e_G & (G\in\mathcal{L}(M))\\ e_G+e_a & (G\notin\mathcal{L}(M))\end{cases}$$

- φ is a graph map of a piecewise-linear function on $\Sigma_{M\setminus a}$

Def

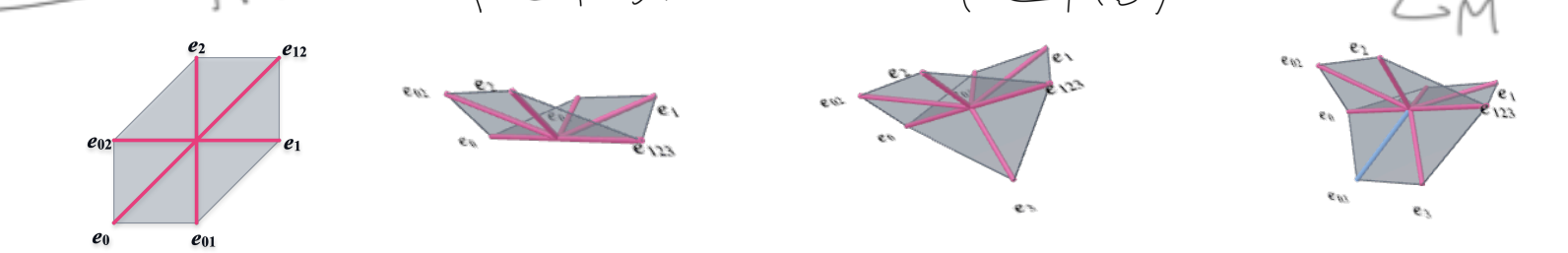
$TM_\varphi(\Sigma_{M\setminus a})$ ← graph of p.-lin. fct. on $\Sigma_{M\setminus a}$

$:=\{\varphi(\sigma) : \sigma\in\Sigma_{M\setminus a}\}\cup\{\varphi(\sigma)+\mathbb{R}_{\ge 0}e_a : \sigma\in\Sigma_{M/a}\}\underset{\text{fan}}{\subset}N_{E,\mathbb{R}}$

called a tropical modification of $\Sigma_{M\setminus a}$ along $\Sigma_{M/a}$

- Bergman fans Σ_M are tropical varieties
- $\varphi(\Sigma_{M\setminus a})$ (=left sum of $TM_\varphi(\Sigma_{M\setminus a})$) is not tropical
- $TM_\varphi(\Sigma_{M\setminus a})$ is the tropical closure of $\varphi(\Sigma_{M\setminus a})$
- (FACT, AP'20) $\pi: TM_\varphi(\Sigma_{M\setminus a})\to\Sigma_{M\setminus a}$ induces
  $\pi^*: A(M\setminus a)\xrightarrow{\sim}A(X(TM_\varphi(\Sigma_{M\setminus a})))$
- (FACT) $|TM_\varphi(\Sigma_{M\setminus a})|=|\Sigma_M|$

Exa. $\Sigma_{M\setminus 3}\overset{\text{graph}}{\rightsquigarrow}\varphi(\Sigma_{M\setminus 3})\subset TM_\varphi(\Sigma_{M\setminus 3})\xleftarrow{\text{star subdiv.}}\Sigma_M$



(II) Blowups

Prop

$TM_\varphi(\Sigma_{M\setminus a})=:\Sigma^{(0)}\leftarrow\Sigma^{(1)}\leftarrow\cdots\leftarrow\Sigma^{(m)}=\Sigma_M$

Star subdivision at a 2-dim'l cone
(blowup at a torus inv. subspace of codim 2)

- Blowup center of $X(\Sigma^{(k)})\longrightarrow X(\Sigma^{(k-1)})$
  $\cong X(\Sigma_{[\emptyset,F_k]_M})\times X(\Sigma_{[F_k\cup\{a\},E]_M})$ ← interval in $\mathcal{L}(M)$
  for some $F_k\in\mathcal{L}(M)$ with $\begin{cases} a\notin F_k, \\ F_k\cup\{a\}\in\mathcal{L}(M)\end{cases}$

Inductive hypothesis on $n=|E|$

$\rightsquigarrow\begin{cases} A^\bullet(X(TM_\varphi(\Sigma_{M\setminus a})))\cong A^\bullet(M\setminus a) : \text{PD of dim } r \\ A^\bullet(\text{Bl cent})\cong A^\bullet([\emptyset,F_k]_M)\otimes A^\bullet([F_k\cup\{a\},E]_M) : \text{PD of dim } r-2\end{cases}$

(III) Consequence of Keel's lemma.

In general, We consider

$$\begin{tikzcd}
\text{excep div} \arrow[r] & E \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\square"] & \tilde{X} \arrow[d] & \text{Blowup of } X \text{ along } Z \arrow[l] \\
\text{closed subvar of codim}=c \arrow[r] & Z \arrow[r, hook, "i"] & X & \text{smooth variety} \arrow[l]
\end{tikzcd}$$

if $\begin{cases} \bullet\ i^*: A^\bullet(X)\to A^\bullet(Z) \text{ is surjective} \\ \bullet\ [Z]\cdot A^{r-c}(X)=A^r(X) \\ \bullet\ A^\bullet(X) \text{ satisfies PD of dim } r \\ \bullet\ A^\bullet(Z) \text{ satisfies PD of dim } r-c \end{cases}$

then $A^\bullet(\tilde{X})$ satisfies PD of dim r

$TM_\varphi(\Sigma_{M\setminus a})=\Sigma^{(0)}\leftarrow\cdots\leftarrow\Sigma^{(m)}=\Sigma_M$

- $X(\Sigma^{(k-1)})\leftarrow X(\Sigma^{(k)})$: Blup at torus inv. subsp of codim 2
- $A(X(\Sigma^{(k-1)}))\twoheadrightarrow A\left(\begin{array}{c}\text{Blow up center of}\\ X(\Sigma^{(k-1)})\leftarrow X(\Sigma^{(k)})\end{array}\right)$
  ($\because$ these are toric variety and torus inv subspace)
- (fact, AP'20) Λ : unimod. fan with $|\Lambda|=|\Sigma_M|$
  $\forall\sigma\in\Lambda_r$ $A^r(X(\Lambda))=\mathbb{Z}[V(\sigma)]$
  $\rightsquigarrow A^r(X(\Sigma^{(k-1)}))=[\text{Bl cent}]A^{r-2}(X(\Sigma^{(k-1)}))$
- $A^\bullet(X(\Sigma^{(k-1)}))$: PD of dim r
- $A^\bullet\left(\begin{array}{c}\text{Blowup center of}\\ X(\Sigma^{(k-1)})\leftarrow X(\Sigma^{(k)})\end{array}\right)$: PD of dim $r-2$

$\overset{\text{Keel}}{\rightsquigarrow}$ $A^\bullet(\Sigma^{(k)})$: PD of dim r

$\overset{\text{induction}}{\rightsquigarrow}$ $A^\bullet(M)$ satisfies PD of dim r //

Appendix

Keel's lemma (1972

$$\begin{tikzcd}
\text{excep. div.} \arrow[r] & E \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\square"] & \tilde{X} \arrow[d, "\pi^*"] & \text{Blowup of } X \text{ along } Z \arrow[l] \\
\text{closed subvar of codim}=c \arrow[r] & Z \arrow[r, hook, "i"] & X & \text{smooth variety} \arrow[l]
\end{tikzcd}$$

If $i^*: A^\bullet(X)\longrightarrow A^\bullet(Z)$ is surjective $\left(\dfrac{A^\bullet(X)}{\mathrm{Ker}\, i^*}\cong A^\bullet(Z)\right)$,

then $\tilde{\pi}^*: A^\bullet(X)[T]\longrightarrow A^\bullet(\tilde{X})$, $T\longmapsto -[E]$

induces an isom. $\dfrac{A^\bullet(X)[T]}{(P(T),\ T\cdot\mathrm{Ker}\, i^*)}\xrightarrow{\sim}A^\bullet(\tilde{X})$

monic polynomial of degree c, costant term $=[Z]$

Taking pull back of coefficients,

$i^*P(T)=T^c+c_1(N_ZX)T^{c-1}+\cdots+c_c(N_ZX)\in A^\bullet(Z)[T]$

Chern polynomial of N_ZX

obs

$\forall\ell\in\mathbb{Z}_{\ge 0}$

$$A^\ell(\tilde{X})\cong\frac{A^\ell(X)\oplus T\cdot A^{\ell-1}(X)\oplus\cdots\oplus T^{c-1}\cdot A^{\ell-c+1}(X)}{(T\cdot\mathrm{Ker}(i^*))_\ell}$$

$$\cong A^\ell(X)\oplus T\cdot A^{\ell-1}(Z)\oplus\cdots\oplus T^{c-1}\cdot A^{\ell-c+1}(Z)$$

Assume $\begin{cases} A^\bullet(X) \text{ satisfies PD of dim } r \\ A^\bullet(Z) \text{ satisfies PD of dim } r-c\end{cases}$

Aim: $A^\bullet(\tilde{X})$ satisfies PD of dim r

- $\ell>r\Longrightarrow\begin{cases} A^\ell(X)=0 \\ A^{\ell-1}(Z)=\cdots=A^{\ell-c+1}(Z)=0 \end{cases}$ ← $\ell-c+1>r-c$
  $\rightsquigarrow A^\ell(\tilde{X})=0$ by
- $\ell=r\Longrightarrow A^r(\tilde{X})\cong A^r(X)\cong\mathbb{Z}$

Assume that $\ell<r$

$A^\ell(\tilde{X})$, $A^{r-\ell}(\tilde{X})$ are finitely generated free abelian group by PD of $A(X)$ and $A(Z)$ and

Consider the matrix of

$A^\ell(\tilde{X})\times A^{r-\ell}(\tilde{X})\longrightarrow A^r(\tilde{X})\cong\mathbb{Z}$

$$\begin{array}{c|cccccc}
 & A^{r-\ell}(X) & A^{r-\ell-1}(Z) & \cdots & & & A^{r-\ell-c+1}(Z)\\
\hline
A^\ell(X) & P_0 & 0 & \cdots & & & 0\\
A^{\ell-1}(Z) & 0 & 0 & \cdots & & 0 & P_1\\
 & \vdots & 0 & \cdots & 0 & P_2 & *\\
 & \vdots & \vdots & & \vdots & & \vdots\\
A^{\ell-c+1}(Z) & 0 & P_{c-1} & * & \cdots & & *
\end{array}$$

$\overset{\det}{\rightsquigarrow}\pm\det P_0\cdot\det P_1\cdots\det P_{c-1}=\pm 1$ //

$\begin{cases} T\cdot\alpha\in T\cdot A^{\ell-1}(X) \\ T^{c-1}\cdot\beta\in T^{c-1}A^{r-\ell-c+1}(X)\end{cases}$

$(T\cdot\alpha)(T^{c-1}\cdot\beta)=T^c\alpha\beta\in T^cA^{r-c}(X)$

$\equiv$ mod $T\cdot\mathrm{Ker}\, i^*$

$-[Z]\alpha\beta\in A^r(X)$

$\updownarrow$ $\quad\cong$

$\alpha\beta\quad A^{r-c}(Z)$